



Karnaugh Maps

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Karnaugh Maps

- A Kmap is a matrix consisting of rows and columns that represent the output values of a Boolean function.
- The output values placed in each cell are derived from the minterms of a Boolean function.
- A *minterm* is a product term that contains all of the function's variables exactly once, either complemented or not complemented.

Description of Kmaps

- For example, the minterms for a function having the inputs x and y are:
- Consider the Boolean function,
- Its minterms are:

Minterm	X	Y
$\bar{x}\bar{y}$	0	0
$\bar{x}y$	0	1
$x\bar{y}$	1	0
xy	1	1

Description of Kmaps

- A Kmap has a cell for each minterm.
- This means that it has a cell for each line for the truth table of a function.
- The truth table for the function $F(x,y) = xy$ is shown at the right along with its corresponding Kmap.

$$F(X, Y) = XY$$

X	Y	XY
0	0	0
0	1	0
1	0	0
1	1	1

X \ Y	0	1
	0	1
0	0	0
1	0	1

Re-arranging the truth table

x	y	minterm
0	0	$x'y'$
0	1	$x'y$
1	0	xy'
1	1	xy



		y	
		0	1
x	0	$x'y'$	$x'y$
	1	xy'	xy

		y	
		0	1
x	0	$x'y'$	$x'y$
	1	xy'	xy

	y'	y
x'	$x'y'$	$x'y$
x	xy'	xy

Karnaugh map simplifications

- Imagine a two-variable sum of minterms:

$$x'y' + x'y$$

- Both of these minterms appear in the top row of a Karnaugh map, which means that they both contain the literal x' .

		y
	$x'y'$	$x'y$
x	xy'	xy

$$\begin{aligned}x'y' + x'y &= x'(y' + y) \\ &= x' \bullet 1 \\ &= x'\end{aligned}$$

$$\begin{aligned}&[\text{Distributive}] \\ &[y + y' = 1] \\ &[x \bullet 1 = x]\end{aligned}$$

More two-variable examples

- Another example expression is $x'y + xy$.
 - Both minterms appear in the right side, where y is uncomplemented.
 - Thus, we can reduce $x'y + xy$ to just y .

		y
	$x'y'$	$x'y$
x	xy'	xy

More two-variable examples

- How about $x'y' + x'y + xy$?
 - We have $x'y' + x'y$ in the top row, corresponding to x' .
 - There's also $x'y + xy$ in the right side, corresponding to y .
 - This whole expression can be reduced to $x' + y$.

			y
	$x'y'$	$x'y$	
x	xy'	xy	

example

		<i>y</i>	
		0	1
<i>x</i>	0	0	1
	1	1	1

$$F = m_0 + m_2 + m_3 = x'y + xy' + xy$$

$$F = xy' + y = x + y$$

A three-variable Karnaugh map

- For a three-variable expression with inputs x , y , z , the arrangement of minterms is more tricky:

		YZ			
		00	01	11	10
X	0	$x'y'z'$	$x'y'z$	$x'yz$	$x'yz'$
	1	$xy'z'$	$xy'z$	xyz	xyz'

		YZ			
		00	01	11	10
X	0	m_0	m_1	m_3	m_2
	1	m_4	m_5	m_7	m_6

A three-variable Karnaugh map

- Another way to label the K-map (use whichever you like):

	y			
	$x'y'z'$	$x'y'z$	$x'yz$	$x'yz'$
X	$xy'z'$	$xy'z$	xyz	xyz'
	z			

	y			
	m_0	m_1	m_3	m_2
X	m_4	m_5	m_7	m_6
	z			

Why the funny ordering?

- With this ordering, any group of 2, 4 or 8 adjacent squares on the map contains common literals that can be factored out.

				y
	$x'y'z'$	$x'y'z$	$x'yz$	$x'yz'$
X	$xy'z'$	$xy'z$	xyz	xyz'
				z

$$\begin{aligned}
 & x'y'z + x'yz \\
 = & x'z(y' + y) \\
 = & x'z \cdot 1 \\
 = & x'z
 \end{aligned}$$

				y
	$x'y'z'$	$x'y'z$	$x'yz$	$x'yz'$
X	$xy'z'$	$xy'z$	xyz	xyz'
				z

$$\begin{aligned}
 & x'y'z' + xy'z' + x'yz' + xyz' \\
 = & z'(x'y' + xy' + x'y + xy) \\
 = & z'(y'(x' + x) + y(x' + x)) \\
 = & z'(y' + y) \\
 = & z'
 \end{aligned}$$

Example K-map simplification

- Let's consider simplifying

$$f(x,y,z) = xy + y'z + xz.$$

- First, you should convert the expression into a sum of minterms form, if it's not already.
 - The easiest way to do this is to make a truth table
 - You can either write out the literals or use the minterm shorthand.
- Here is the truth table and sum of minterms for our example:

Example K-map simplification

$$f(x,y,z) = xy + y'z + xz.$$

- Here is the truth table and sum of minterms for our example:

x	y	z	$f(x,y,z)$
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$$\begin{aligned} f(x,y,z) &= x'y'z + xy'z + xyz' + xyz \\ &= m_1 + m_5 + m_6 + m_7 \end{aligned}$$

Making the example K-map

$$f(x,y,z) = x'y'z + xy'z + xyz' + xyz$$

$$f(x,y,z) = m_1 + m_5 + m_6 + m_7$$

	y			
	$x'y'z'$	$x'y'z$	$x'yz$	$x'yz'$
X	$xy'z'$	$xy'z$	xyz	xyz'
	z			

	y			
	m_0	m_1	m_3	m_2
X	m_4	m_5	m_7	m_6
	z			

	y			
	0	1	0	0
X	0	1	1	1
	z			

Simplify the K-map

- Each rectangle corresponds to one product term.
- The product is determined by finding the common literals in that rectangle.

			y	
	0	1	0	0
x	0	1	1	1
		z		

			y	
	$x'y'z'$	$x'y'z$	$x'yz$	$x'yz'$
x	$xy'z'$	$xy'z$	xyz	xyz'
		z		

For our example, we find that

$$xy + y'z + xz = y'z + xy.$$

Practice K-map I

- Simplify the sum of minterms $m_1 + m_3 + m_5 + m_6$.

The image shows two Karnaugh maps for a 4-variable function. The top map has variables X, Y, Z, and Y (labeled as such). The bottom map has variables X, Y, Z, and Y (labeled as such). The bottom map has a pink box around the cell (X=1, Y=1, Z=1, Y=1) and a blue box around the cell (X=0, Y=1, Z=1, Y=1).

- Minterm m_6 is not in a group.
- The final MSP here is $x'z + y'z + xyz$.

example

- Simplify the Boolean function

$$F(x, y, z) = \sum (2, 3, 4, 5)$$

Answer:

		yz			
		00	01	11	10
X	0			1	1
	1	1	1		

- $m_2 + m_3 = x'y$

- $m_4 + m_5 = xy'$

- So, the simplified Boolean function is :

- $F = xy' + x'y$

example

- Simplify the Boolean function

$$F(x, y, z) = \sum (0, 2, 4, 5, 6)$$

Answer:

A Karnaugh map for the Boolean function $F(x, y, z) = \sum (0, 2, 4, 5, 6)$. The map is a 2x4 grid with columns labeled yz (00, 01, 11, 10) and rows labeled x (0, 1). The cells containing 1s are at (0,00), (0,10), (1,00), (1,01), and (1,10). Three blue groupings are shown: a group of two cells in the top row (0,00) and (0,10); a group of two cells in the bottom row (1,00) and (1,01); and a group of two cells in the rightmost column (0,10) and (1,10).

	yz			
	00	01	11	10
x 0	1			1
1	1	1		1

- $m_0 + m_2 + m_4 + m_6 = z'$

- $m_4 + m_5 = xy'$

- So, the simplified Boolean function is :

- $F = z' + xy'$

example

- Simplify the Boolean function

$$F(x, y, z) = \sum (3, 4, 6, 7)$$

Answer:

A Karnaugh map for the Boolean function $F(x, y, z) = \sum (3, 4, 6, 7)$. The map is a 2x4 grid with columns labeled yz (00, 01, 11, 10) and rows labeled x (0, 1). The cells containing 1s are at (0, 11), (1, 00), (1, 11), and (1, 10). Three blue groupings are shown: a vertical group of two cells at $yz=11$, a horizontal group of two cells at $x=1$ for $yz=00$ and $yz=01$, and a horizontal group of two cells at $x=1$ for $yz=10$ and $yz=11$.

	yz			
	00	01	11	10
x 0			1	
1	1		1	1

- $m_3 + m_7 = yz$

- $m_4 + m_6 = xz'$

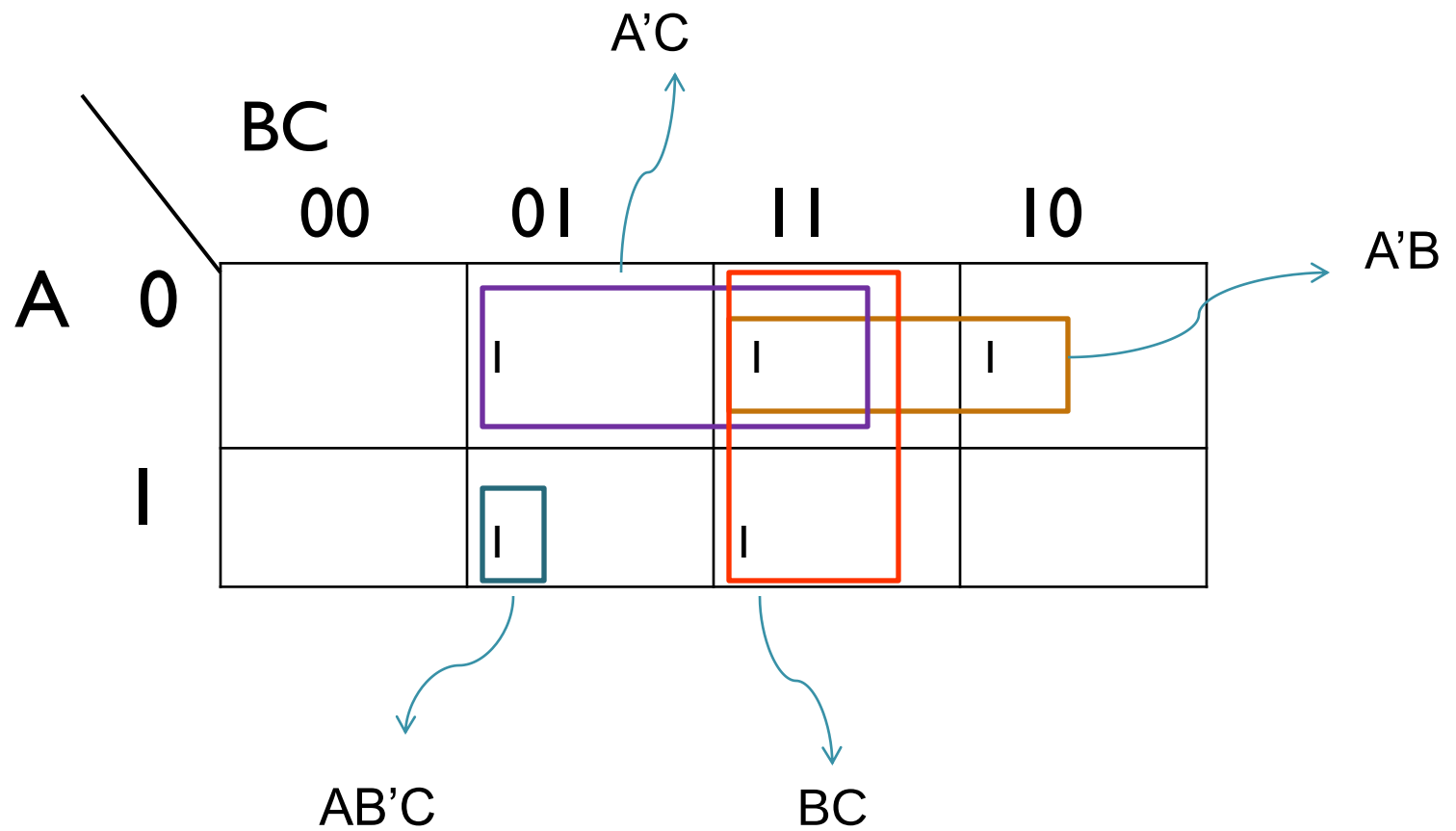
- So, the simplified Boolean function is :

- $F = yz + xz'$

example

- For the Boolean function
- $F = A'C + A'B + AB'C + BC$
- i. Express this function as sum of minterms
- ii. Find the minimal sum of products expression

Answer



Answer:

i. $F(A, B, C) = \Sigma(1,2,3,5,7)$

ii. $m_1 + m_3 + m_5 + m_7 = C$

$$m_2 + m_3 = A'B$$

$$F = C + A'B$$

Three-Variable K-Maps

$$f = \sum(0,4) = \bar{B} \bar{C}$$

A \ BC				
	00	01	11	10
0	1	0	0	0
1	1	0	0	0

$$f = \sum(4,5) = A \bar{B}$$

A \ BC				
	00	01	11	10
0	0	0	0	0
1	1	1	0	0

$$f = \sum(0,1,4,5) = \bar{B}$$

A \ BC				
	00	01	11	10
0	1	1	0	0
1	1	1	0	0

$$f = \sum(0,4) = \bar{A} C$$

A \ BC				
	00	01	11	10
0	0	1	1	0
1	0	0	0	0

$$f = \sum(4,6) = A \bar{C}$$

A \ BC				
	00	01	11	10
0	0	0	0	0
1	1	0	0	1

$$f = \sum(0,2) = \bar{A} \bar{C}$$

A \ BC				
	00	01	11	10
0	1	0	0	1
1	0	0	0	0

Four-variable K-maps

- We can do four-variable expressions too!
 - The minterms in the third and fourth columns, *and* in the third and fourth rows, are switched around.
 - Again, this ensures that adjacent squares have common literals.
- Grouping minterms is similar to the three-variable case, but:
 - You can have rectangular groups of 1, 2, 4, 8 or 16 minterms.
 - You can wrap around *all four* sides.

Four-variable K-maps

			y			
			w'x'y'z'	w'x'y'z	w'x'yz	w'x'yz'
			w'xy'z'	w'xy'z	w'xyz	w'xyz'
W		wxy'z'	wxy'z	wxyz	wxyz'	X
		wx'y'z'	wx'y'z	wx'yz	wx'yz'	
			Z			

		y				
		m ₀	m ₁	m ₃	m ₂	
		m ₄	m ₅	m ₇	m ₆	X
W		m ₁₂	m ₁₃	m ₁₅	m ₁₄	
		m ₈	m ₉	m ₁₁	m ₁₀	
		Z				

Example

- Simplify

$$F = m_0 + m_2 + m_5 + m_8 + m_{10} + m_{13}$$

- The expression is already a sum of minterms, so here's the K-ma

		y		
w	1	0	0	1
	0	1	0	0
	0	1	0	0
	1	0	0	1
		z		x

		y		
w	m_0	m_1	m_3	m_2
	m_4	m_5	m_7	m_6
	m_{12}	m_{13}	m_{15}	m_{14}
	m_8	m_9	m_{11}	m_{10}
		z		x

Example

- We can make the following groups, resulting in the MSP $x'z' + xy'z$.

			y	
	1	0	0	1
	0	1	0	0
	0	1	0	0
W	1	0	0	1
			z	

			y	
	$w'x'y'z'$	$w'x'y'z$	$w'x'yz$	$w'x'yz'$
	$w'xy'z'$	$w'xy'z$	$w'xyz$	$w'xyz'$
	$wxy'z'$	$wxy'z$	$wxyz$	$wxyz'$
W	$wx'y'z'$	$wx'y'z$	$wx'yz$	$wx'yz'$
			z	

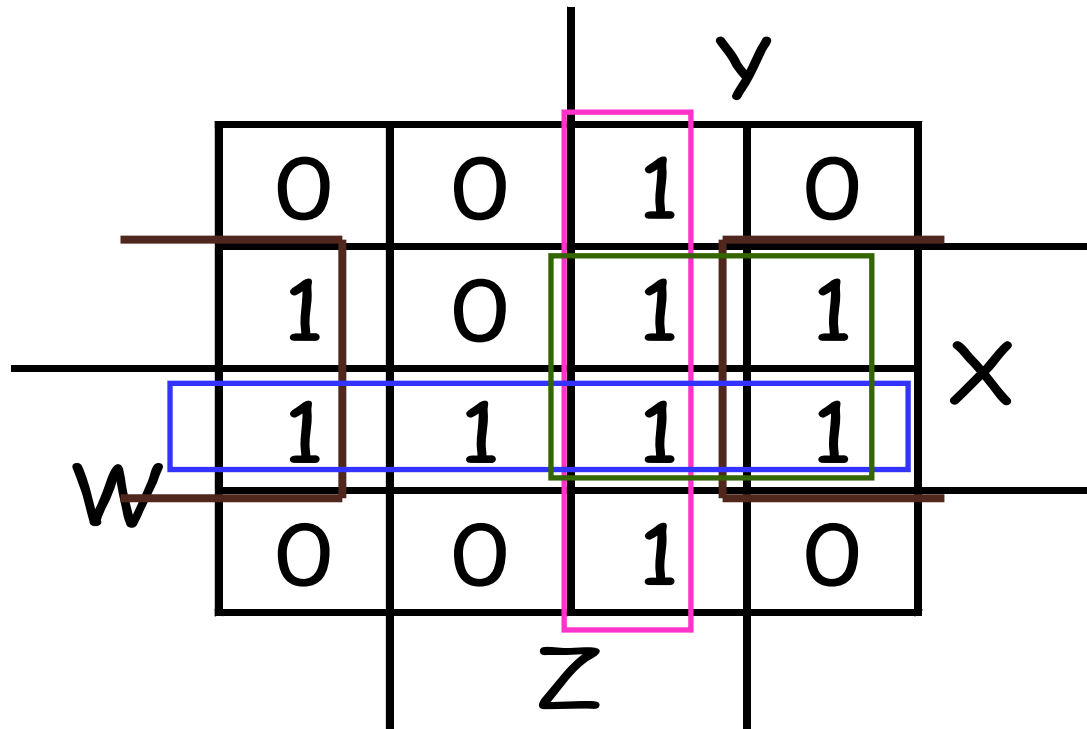
Example

- Simplify for the following K-map:

		y		
		0	1	
w	0	0	1	0
	1	0	1	1
	1	1	1	1
	0	0	1	0
		z		
		0	1	

Example

- Simplify for the following K-map:



example

- Simplify the Boolean function
- $F = A'B'C' + B'CD' + A'BCD' + AB'C'$

		CD			
		00	01	11	10
AB	00	1	1		1
	01				1
	11				
	10	1	1		1

Answer

- $m_0 + m_1 + m_8 + m_9 = B'C'$
- $m_0 + m_2 + m_8 + m_{10} = B'D'$
- $m_2 + m_6 = A'CD'$
- $F = B'C' + B'D' + A'CD'$

Example

AB \ CD	CD			
	00	01	11	10
00	1	0	0	0
01	0	0	0	0
11	0	0	0	0
10	1	0	0	0

$$f = \sum (0,8) = \bar{B} \bullet \bar{C} \bullet \bar{D}$$

AB \ CD	CD			
	00	01	11	10
00	0	0	0	0
01	0	1	0	0
11	0	1	0	0
10	0	0	0	0

$$f = \sum (5,13) = B \bullet \bar{C} \bullet D$$

AB \ CD	CD			
	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	1	1	0
10	0	0	0	0

$$f = \sum (13,15) = A \bullet B \bullet D$$

AB \ CD	CD			
	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	0	0	0	0
10	0	0	0	0

$$f = \sum (2,3,6,7) = \bar{A} \bullet C$$

AB \ CD	CD			
	00	01	11	10
00	0	0	0	0
01	1	0	0	1
11	1	0	0	1
10	0	0	0	0

$$f = \sum (4,6,12,14) = B \bullet \bar{D}$$

AB \ CD	CD			
	00	01	11	10
00	0	0	1	1
01	0	0	0	0
11	0	0	0	0
10	0	0	1	1

$$f = \sum (2,3,10,11) = \bar{B} \bullet C$$

Example

		CD			
		00	01	11	10
AB	00	0	0	0	0
	01	1	1	1	1
	11	0	0	0	0
	10	0	0	0	0

$$f = \Sigma(4,5,6,7) = \bar{A} \bullet B$$

		CD			
		00	01	11	10
AB	00	0	0	1	0
	01	0	0	1	0
	11	0	0	1	0
	10	0	0	1	0

$$f = \Sigma(3,7,11,15) = C \bullet D$$

		CD			
		00	01	11	10
AB	00	1	0	1	0
	01	0	1	0	1
	11	1	0	1	0
	10	0	1	0	1

$$f = \Sigma(0,3,5,6,9,10,12,15)$$

$$f = A \otimes B \otimes C \otimes D$$

		CD			
		00	01	11	10
AB	00	0	1	1	0
	01	0	1	1	0
	11	0	1	1	0
	10	0	1	1	0

$$f = \Sigma(1,3,5,7,9,11,13,15)$$

$$f = D$$

		CD			
		00	01	11	10
AB	00	1	0	0	1
	01	1	0	0	1
	11	1	0	0	1
	10	1	0	0	1

$$f = \Sigma(0,2,4,6,8,10,12,14)$$

$$f = \bar{D}$$

		CD			
		00	01	11	10
AB	00	0	0	0	0
	01	1	1	1	1
	11	1	1	1	1
	10	0	0	0	0

$$f = \Sigma(4,5,6,7,12,13,14,15)$$

$$f = B$$